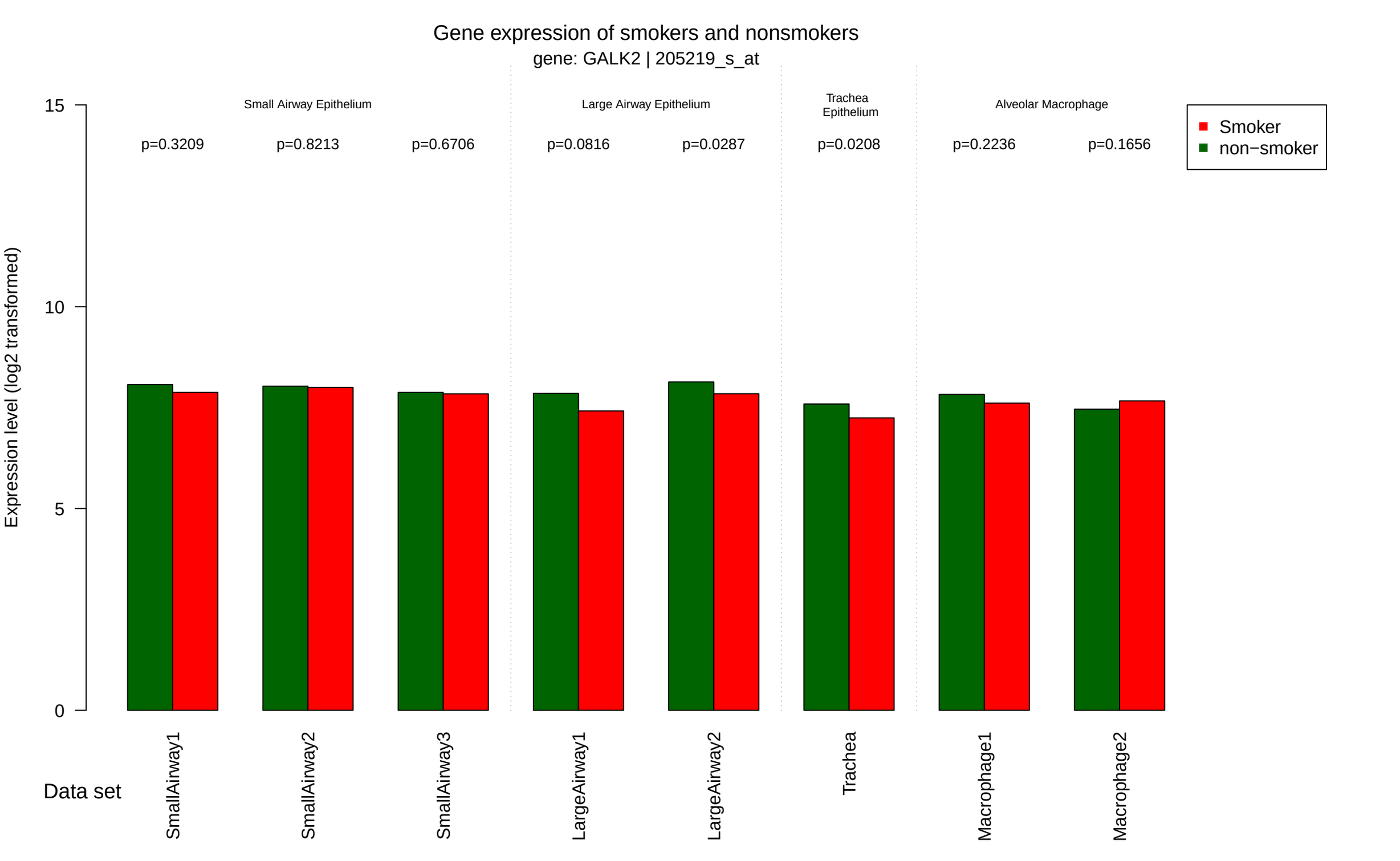
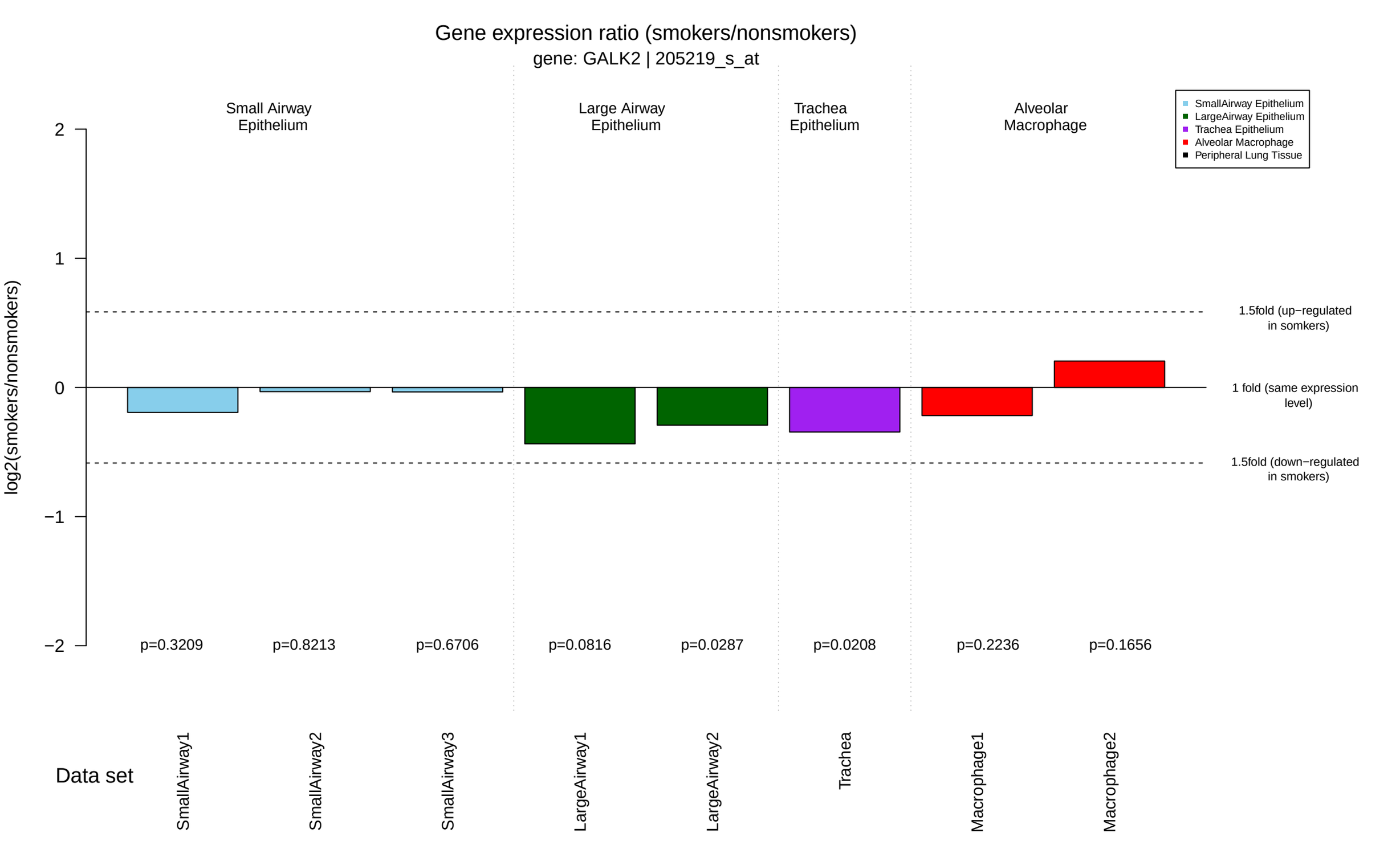
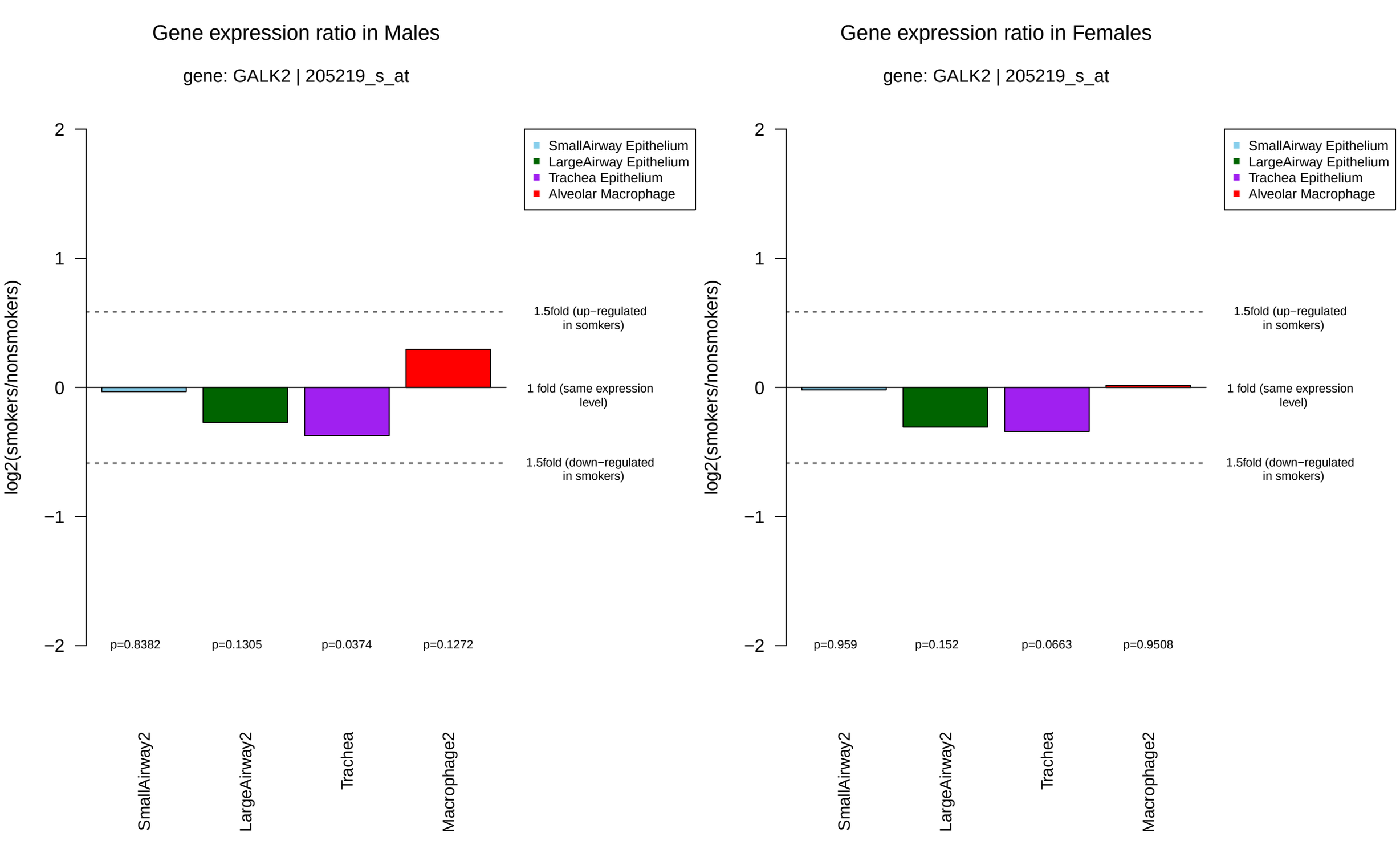
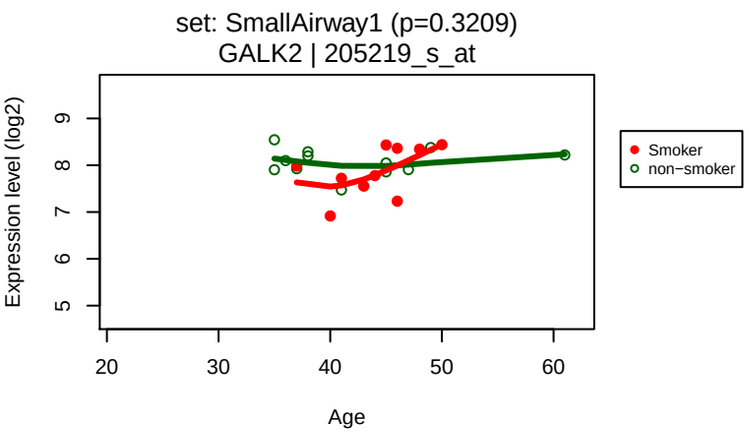
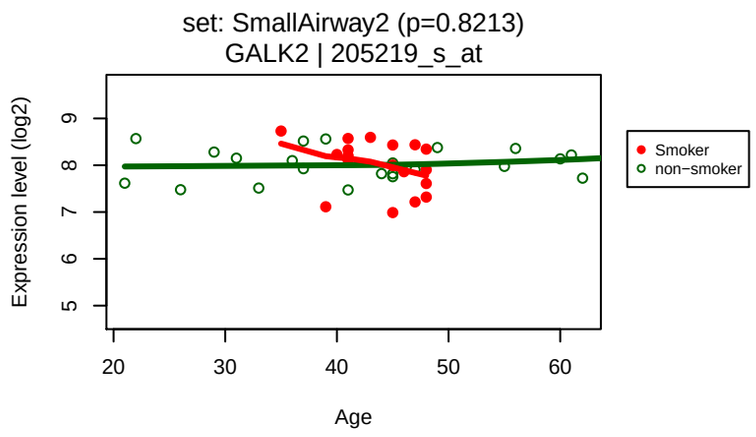
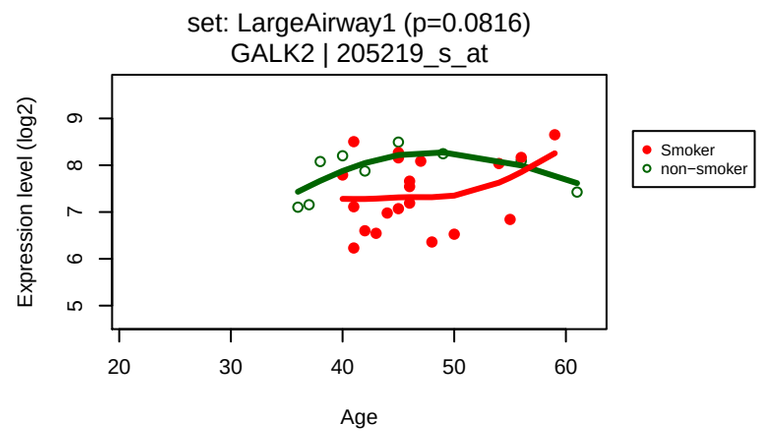

Gene expression of smokers and nonsmokers
gene: GALK2 | 205219_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3209
p=0.8213
p=0.6706
p=0.0816
p=0.0287
p=0.0208
p=0.2236
p=0.1656
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

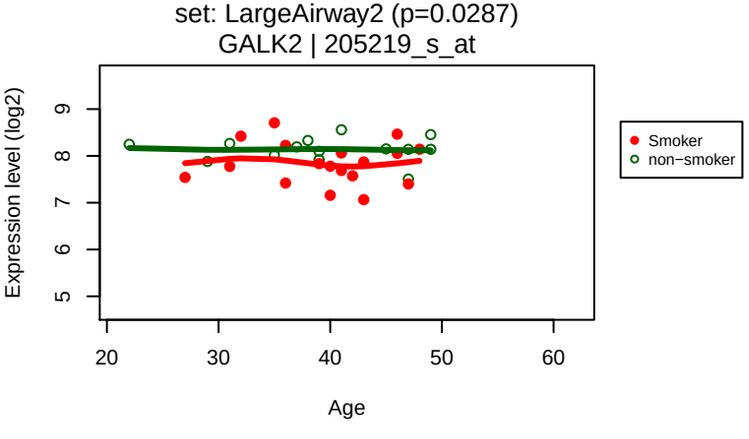

Gene expression ratio (smokers/nonsmokers)
gene: GALK2 | 205219_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3209
p=0.8213
p=0.6706
p=0.0816
p=0.0287
p=0.0208
p=0.2236
p=0.1656
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

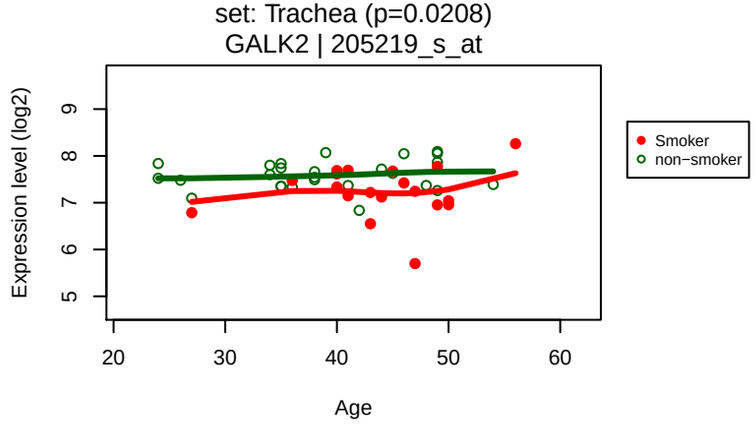

Gene expression ratio in Males
gene: GALK2 | 205219_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8382
p=0.1305
p=0.0374
p=0.1272
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GALK2 | 205219_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.959
p=0.152
p=0.0663
p=0.9508
SmallAirway2
LargeAirway2
Trachea
Macrophage2

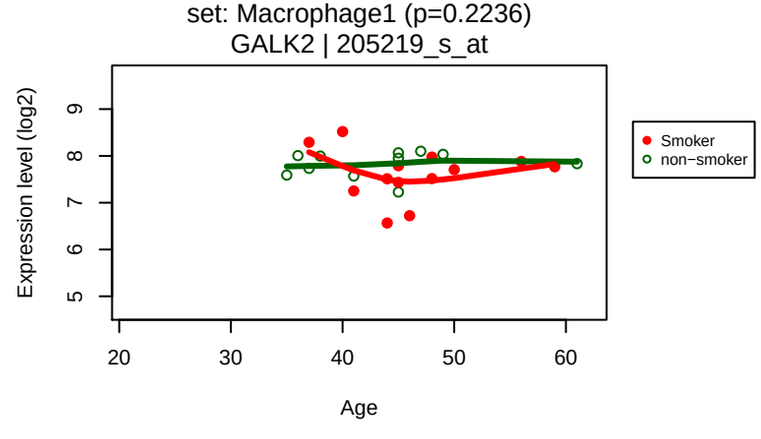

set: SmallAirway1 (p=0.3209)
GALK2 | 205219_s_at
Expression level (log2)
Age
Smoker
non−smoker

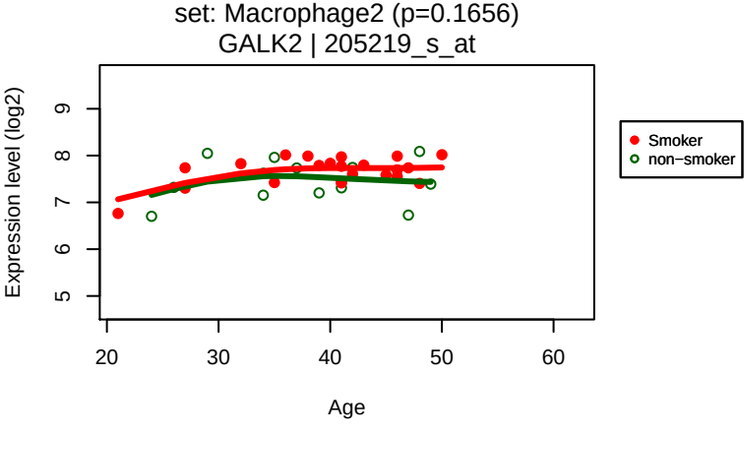

set: SmallAirway2 (p=0.8213)
GALK2 | 205219_s_at
Expression level (log2)
Age
Smoker
non−smoker

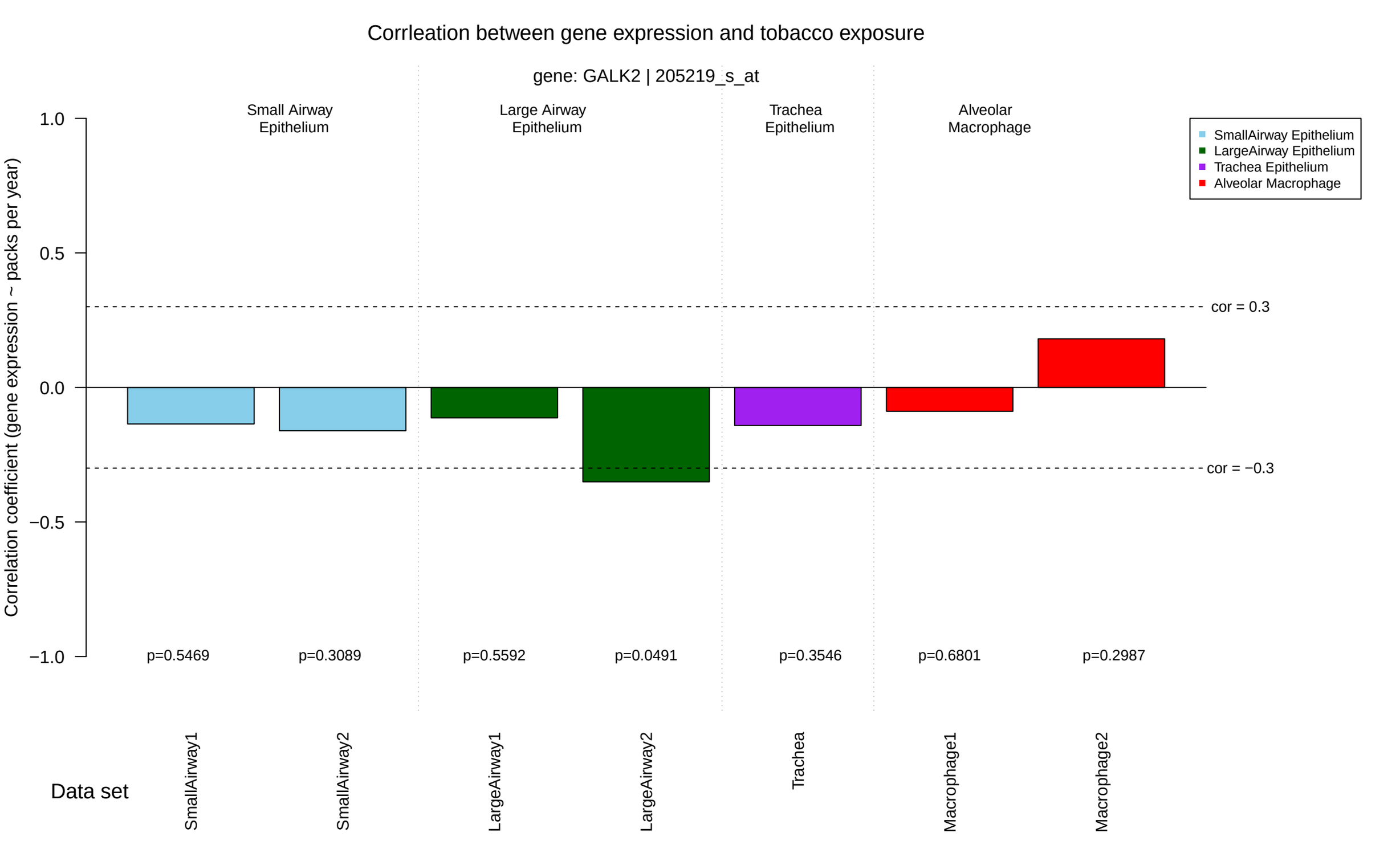

set: LargeAirway1 (p=0.0816)
GALK2 | 205219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0287)
GALK2 | 205219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0208)
GALK2 | 205219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2236)
GALK2 | 205219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1656)
GALK2 | 205219_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GALK2 | 205219_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5469
p=0.3089
p=0.5592
p=0.0491
p=0.3546
p=0.6801
p=0.2987
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2